Announced Rescue Pricing with Latent Market Thickness

Group-meeting note — benchmark, numerical pattern, and formal model

27 August 2026

Preview. A publicly known single driver gives an exact cutoff-WPBE and a closed-form optimal menu. With latent Poisson supply, the completion value of rescue pricing is small in thin markets, largest at an intermediate thickness, and small again in thick markets. For $(\alpha, \beta, \delta, \gamma) = (1, .8, .8, 0)$, the deterministic numerical solution peaks near $m = 9.72$ and improves optimized completion by 6.16 percentage points. The endpoint pattern is proved; strict single-peakedness is a conjecture.

1 Exact one-rider/one-driver benchmark

1.1 Closed-form solution

The rider has $v \sim U[0, 1]$, the driver has $c \sim U[0, 1]$, and the platform maximizes completion. Delayed rider value is βv and delayed incumbent surplus is $\delta(p_j - c)$. The platform commits to $0 \le p_1 \le p_2 \le 1$. The number of drivers is publicly known to be exactly one; there is no survival shock or fresh entry.

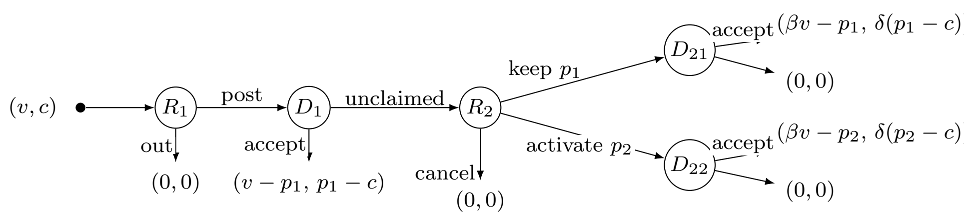

Figure 1: Extensive form.

For an active menu $0 \le p_1 < p_2 < \beta$, the driver accepts immediately iff $c \le \widehat{c}$. After rejection, the rider cancels, repeats p_1, or activates p_2 according to

$$v < \frac{p_1}{\beta}, \qquad \frac{p_1}{\beta} \le v \le v^M(\widehat{c}), \qquad v > v^M(\widehat{c}), \qquad v^M(\widehat{c}) = \frac{p_1 + p_2 - \widehat{c}}{\beta}.$$

The marginal driver's accept-versus-wait indifference gives the WPBE cutoff

$$\widehat{c}(p_1, p_2) = \left[p_1 - \frac{\delta(p_2 - p_1)(\beta - p_2)}{\beta(1 - p_1) - \delta(\beta - p_2)}\right]_+, \tag{1}$$

and equilibrium completion is

$$M_1(p_1, p_2) = (1 - p_1)\widehat{c} + \frac{(\beta - p_2)(p_2 - \widehat{c})}{\beta}. \tag{2}$$

Thus the cutoff is calculated from the WPBE; the platform then optimizes over menus subject to this equilibrium response.

Fixing a target cutoff reduces the two-price problem to one dimension. The unique continuation envelope and its implementing first price are

$$\widehat{p}_2(\widehat{c}) = \frac{\beta + \widehat{c}}{2}, \qquad \widehat{p}_1(\widehat{c}) = \frac{1 + \widehat{c} - \sqrt{\Delta(\widehat{c})}}{2}, \qquad \Delta(\widehat{c}) = (1 - \widehat{c})^2 - \frac{\delta}{\beta}(\beta - \widehat{c})^2. \tag{3}$$

Substitution into (2) gives

$$M_1^* = \max\left\{\frac{1}{4},\ \max_{\widehat{c} \in [0,\beta]} J_\delta(\widehat{c})\right\}, \quad J_\delta(\widehat{c}) = \frac{\widehat{c}[1 - \widehat{c} + \sqrt{\Delta(\widehat{c})}]}{2} + \frac{(\beta - \widehat{c})^2}{4\beta}. \tag{4}$$

If $\beta \le 1/2$ or $\delta = 1$, the optimal value is the flat benchmark $1/4$. If $\beta > 1/2$ and $\delta < 1$, J_δ is strictly concave and has a unique interior maximizer, which implements $p_1^* < p_2^*$.

1.2 Numerical illustration

At $\beta = \delta = 4/5$, the first-order condition becomes $12 - 20\widehat{c} = 5\widehat{c}\sqrt{9 - 10\widehat{c}}$. Squaring yields $(5\widehat{c} - 4)(50\widehat{c}^2 + 75\widehat{c} - 36) = 0$; checking the unsquared equation and feasibility leaves the positive root of the quadratic. Its discriminant is $75^2 + 4(50)(36) = 225 \cdot 57$, which explains the appearance of $\sqrt{57}$:

$$\widehat{c}^* = \frac{3(\sqrt{57} - 5)}{20} = .382475, \quad p_1^* = \frac{11 + \sqrt{57}}{40} = .463746, \quad p_2^* = \frac{1 + 3\sqrt{57}}{40} = .591238.$$

The optimal rescue menu completes 25.9581% of requests, compared with 25.0000% under the best flat price.

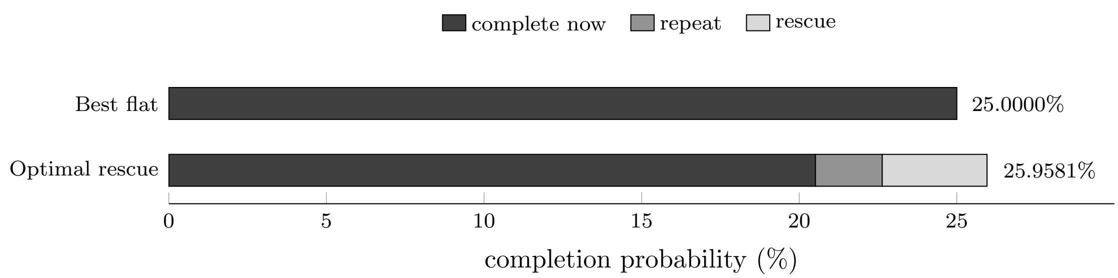

Figure 2: Exact-one-driver completion at $\beta = \delta = .8$. The 0.9581 pp gain combines less completion now with repeat and rescue completion later.

β	δ	$\widehat{c}^*$	p_1^*	p_2^*	M_1^*
.6	.5	.460069	.467733	.530034	.253038
.6	.8	.478613	.488210	.539306	.251089
.8	.5	.370421	.420148	.585210	.272458
.8	.8	.382475	.463746	.591238	.259581
.9	.5	.329857	.405843	.614928	.286282
.9	.8	.314018	.453693	.607009	.266932

Each row re-solves the WPBE and optimal menu. More patient riders support a larger rescue region; less patient incumbents make waiting less attractive and increase the completion value of escalation.

2 Why the design becomes (p_1, p_2, s)

Takeaway. Pages 1–2 isolate same-order strategic waiting in the exact one-driver benchmark. The market-design comparison now adds a fresh Poisson cohort in every second window and asks whether the platform should reprice the same catchment, widen the catchment, or do both.

2.1 Three strictly nested mechanism classes

Mechanism class	Commitment	Second-window core	What is added?
Flat fixed reach $\mathcal{M}_0$	$(p, p, 1)$	$N_2^C \sim \text{Pois}(m)$	No contingent price and no outer search.
Fixed-reach rescue $\mathcal{M}_F$	$(p_1, p_2, 1)$	$N_2^C \sim \text{Pois}(m)$	A committed rescue price in the same catchment.
Expanded-search rescue $\mathcal{M}_E$	(p_1, p_2, s), $s \geq 1$	$N_2^C \sim \text{Pois}(m)$	An independent outer annulus plus the same rescue price.

Thus "fixed reach" fixes the notification geography, not driver identities. Both rescue mechanisms contain time-homogeneous fresh arrivals in period 2. Their feasible sets are

$$\mathcal{M}_0 = \{(p, p, 1)\} \subset \mathcal{M}_F = \{(p_1, p_2, 1) : p_2 \geq p_1\} \subset \mathcal{M}_E = \{(p_1, p_2, s) : p_2 \geq p_1,\ 1 \leq s \leq \bar{s}\}. \tag{5}$$

The flat benchmark has one optimized price; it is not a fixed-p_1 policy.

2.2 s is an area multiplier, not a participation share

Let spatially homogeneous fresh drivers form a Poisson point process. Normalize the core catchment to area one and let its radius be R_0. Committing to multiplier s means

$$R(s) = R_0\sqrt{s}, \qquad N_2^O(s) \sim \text{Pois}((s-1)m), \qquad N_2^C \perp N_2^O(s). \tag{6}$$

Hence the expanded pool is $\text{Pois}(sm)$ and the core is its Poisson thinning with retention probability $1/s$. At $s = 4$, for example, area is quadrupled but radius only doubles.

2.3 The object plotted below

For market environment $\theta = (m, \beta, \delta)$, each line reports a separate equilibrium-constrained optimum

$$V_k(\theta) = \max_{\mu \in \mathcal{M}_k} \min_{a \in \mathcal{E}^{\text{WPBE}}(\mu;\theta)} M(\mu, a), \qquad k \in \{0, F, E\}. \tag{7}$$

The nesting makes $V_E \geq V_F \geq V_0$; the substantive question is where each increment is large after all permitted controls are re-optimized.

3 Inner cutoff-WPBE, then outer mechanism design

Takeaway. The platform commits first. For every candidate mechanism, driver strategies, rider continuation, and Bayesian beliefs are solved as a complete cutoff-WPBE. Only then does the platform compare mechanisms.

3.1 Information and timing for the same focal order

Stage	Action	Information
0	Platform commits to $\mu = (p_1, p_2, s)$; a rider of value v posts iff $v \geq p_1$.	m, β, δ, the cost laws, and μ are public.
1	$N_1 \sim \mathrm{Pois}(m)$ incumbents see this order and choose accept p_1 or reject-and-wait.	Each sees the order, both prices, s, and own persistent cost c; counts and other costs are hidden.
2	After universal rejection, the rider chooses abandon, repeat at p_1, or rescue at p_2.	The rider sees failure but no driver count or future fresh responses; beliefs are induced by the equilibrium cutoff.
3	The chosen terminal notification is sent; fresh drivers see the order for the first time and may respond.	Rejectors and fresh volunteers enter one anonymous lottery. A rejector may first exit or lose eligibility; pickup cost is paid only by the selected driver.

3.2 Winner-only pickup cost and terminal coverage

Write area rank as $u = (r/R_0)^2$. Core pickup is absorbed into $c \sim U[0,1]$; the extra cost from expansion is

$$d(u) = \tau(\sqrt{u} - 1)_+.$$

A fresh driver's response payoff is $h(\lambda)[p - c - d(u)]$, where $h(z) = (1 - e^{-z})/z$. Because the same assignment probability multiplies payment and pickup cost,

$$\text{respond} \iff c + d(u) \leq p, \qquad e(p, s) = m \int_0^s F(p - d(u))\, du. \tag{8}$$

There is no sunk relocation cost and no fresh-entry fixed point. Let a rejector remain available with probability α and remain eligible conditional on availability with probability χ; write $\omega = \alpha\chi$. Given cutoff a, let $s_1 = 1, s_2 = s$ and

$$I_j^{\omega}(a) = \omega m[F(p_j) - F(a)]_+, \quad \lambda_j(a) = I_j^{\omega}(a) + e(p_j, s_j), \quad C_j(a) = 1 - e^{-\lambda_j(a)}. \tag{9}$$

The rider compares 0, $C_1(\beta v - p_1)$, and $C_2(\beta v - p_2)$; let the resulting conditional action masses be $\eta_0(a), \eta_1(a), \eta_2(a)$.

3.3 Strict cutoff-WPBE test and the outer problem

For a proposed aggregate cutoff, a type-c incumbent's accept-minus-wait payoff is

$$D(c; a) = h(mF(a))(p_1 - c) - \delta e^{-mF(a)} \omega \sum_{j=1}^{2} \eta_j(a) h(\lambda_j(a))(p_j - c)_+. \tag{10}$$

Together with the rider and fresh-driver strategies above, a is a cutoff-WPBE iff beliefs obey Bayes' rule and

$$D(c; a) \geq 0 \ \forall c < a, \qquad D(c; a) \leq 0 \ \forall c > a, \tag{11}$$

with indifference at an interior cutoff. Since D is piecewise affine with kinks only at p_1, p_2, the numerics verify (11) exactly at interval endpoints, not on a sampled type grid. The computation respects the nesting

$$\text{fix } p_1 \longrightarrow \max_{p_2, s} \min_{a \in \mathcal{E}^{\mathrm{WPBE}}} M \longrightarrow \max_{p_1}, \tag{12}$$

and enumerates the entire cutoff correspondence before conservative selection. The archived curves set $\omega = 1$; lower retention is an explicit primitive.

4 Part 2: fixed-reach rescue

Takeaway. Fixed-reach rescue reprices the second window without changing its geography. It still has a new core Poisson cohort; its distinctive tool is the committed wedge $p_2 - p_1$, not the existence of fresh arrivals.

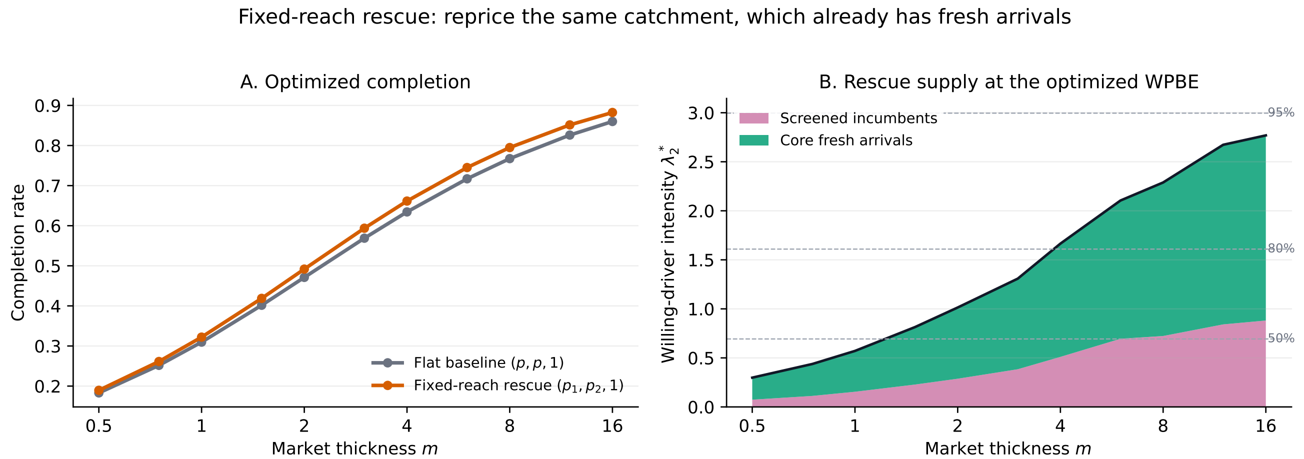

Figure 3: Fixed-reach rescue versus the flat two-window benchmark. Every point first solves the complete cutoff-WPBE and then re-optimizes the permitted mechanism. Right: second-window willing-driver intensity under the optimized fixed-rescue policy.

Under repeat, the terminal pool contains screened incumbents $\omega m[F(p_1) - F(a)]_+$ and core fresh volunteers $e(p_1, 1)$. Under rescue, it contains $\omega m[F(p_2) - F(a)]_+$ and $e(p_2, 1)$. For uniform costs,

$$e(p, 1) = mp. \tag{13}$$

Thus increasing p_2 works through two supply channels, but it also increases the continuation value of rejecting the focal order and may lower a.

At $(\beta, \delta, \tau) = (.8, .8, .25)$, the optimized fixed-rescue increment is only 0.65 percentage points at $m = .5$, peaks at 2.81 points near $m = 6$, and remains 2.24 points at $m = 16$. This hump is an equilibrium-design pattern: each orange point has its own (p_1^*, p_2^*) and its own WPBE cutoff.

Interpretation. In a very thin market, repricing a small catchment has little supply to work with. As m grows, the second-window core cohort and screened backup pool make contingent pay more useful. In a very thick market, the flat mechanism already completes most requests, limiting the remaining gain.

5 Part 3: expanded-search rescue

Takeaway. Expanded search preserves the core cohort and adds only the outer annulus. Distant drivers volunteer less often because the assigned winner, and only the winner, pays the incremental pickup cost.

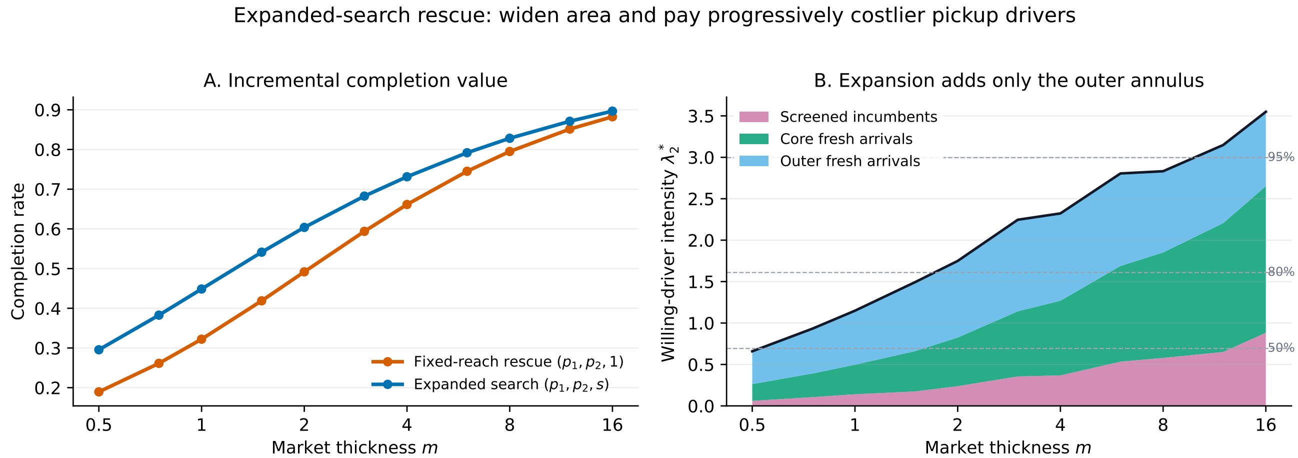

Figure 4: Incremental value and supply composition of expanded search. Each blue point re-optimizes (p_1, p_2, s) outside its complete cutoff-WPBE; the orange comparator separately re-optimizes $(p_1, p_2, 1)$.

For $F = U[0,1]$, $p \in [0,1]$, and $\tau > 0$, define

$$R(p,s) = \min\{\sqrt{s}, 1 + p/\tau\}.$$

Poisson thinning gives the closed-form fresh volunteer intensity

$$e(p,s) = mp + m(p+\tau)(R^2 - 1) - \frac{2m\tau}{3}(R^3 - 1). \tag{14}$$

Its marginal outer supply is

$$\frac{\partial e}{\partial s} = m[p - \tau(\sqrt{s} - 1)]_+, \qquad s^{\text{sat}}(p) = (1 + p/\tau)^2. \tag{15}$$

Thus pickup friction creates a physical response saturation. It does not make notifications economically free. If expanded rescue is executed with ex-ante probability q_R, expected extra contacts are

$$Q^O = q_R m(s-1), \qquad J_\kappa = BM - \kappa Q^O. \tag{16}$$

Completion with $\kappa = 0$ is a maximal-completion frontier; an economic scope optimum requires $\kappa > 0$ or an equivalent notification budget.

At $(\beta, \delta, \tau) = (.8, .8, .25)$, the incremental value of expanded search over optimized fixed rescue peaks at 12.64 percentage points near $m = 1$ and falls to 1.42 points at $m = 16$. The optimized multiplier declines from its cap $s^* = 4$ in thin markets to $s^* = 2.09$ at $m = 16$.

Interpretation. Price chooses which reached drivers are willing; s chooses who can first learn about the failed order. With free contacts, s^* may hit $\bar{s}$ or the physical saturation set. With $\kappa > 0$, search expands only while its marginal completion value exceeds the opportunity cost of another expected contact.

6 The two controls peak in different thickness regions

Takeaway. In the fully optimized cutoff-WPBE numerics, expanded search creates its largest incremental gain first; contingent pricing reaches its maximum later. The right panel gives the common-branch theorem that explains this ordering. The left panel is evidence for the re-optimized model, not its proof.

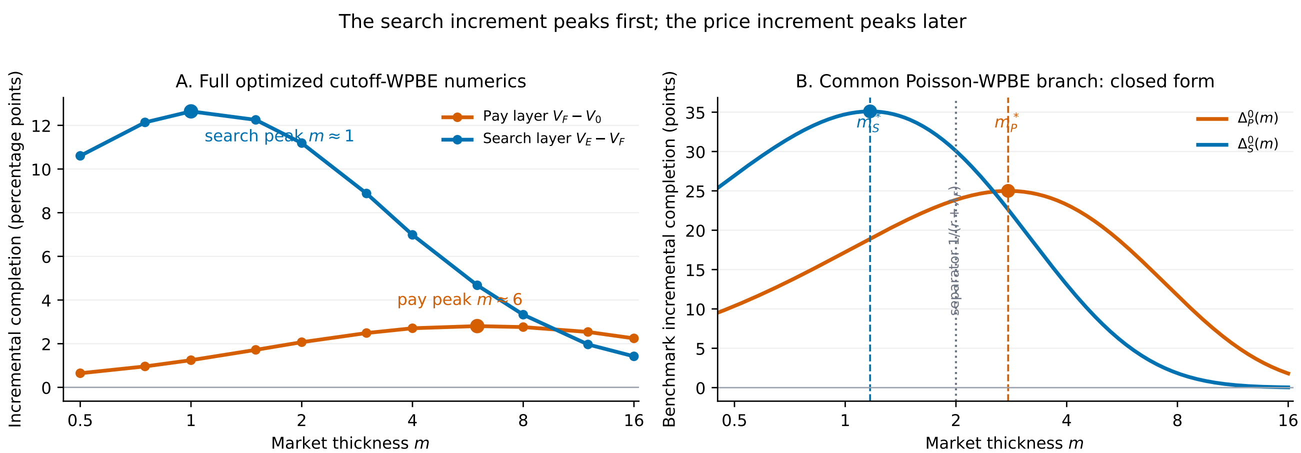

Figure 5: Left: every point separately re-solves the inner cutoff-WPBE and re-optimizes all controls available in that mechanism class. Right: one illustrative set of effective rates for the closed-form theorem; the ordering holds for every $0 \leq \lambda_0 < \lambda_F < \lambda_E$ with $r + \lambda_0 > 0$.

Thickness regime	Expanded-search layer $V_E - V_F$	Price layer $V_F - V_0$
Scarce	Geographic reach is binding; the gain peaks near $m = 1$ in the main slice.	Repricing a small core acts on little supply.
Intermediate	Its marginal value falls as the core itself becomes reliable.	The screened and core-fresh backup pools are now large enough for contingent pay to peak near $m = 6$.
Ample	Added outer coverage has little room to improve completion.	The flat optimized mechanism also becomes reliable, so the price increment declines.

Exact status. The numerical peak locations use $(\beta, \delta, \tau, \omega) = (.8, .8, .25, 1)$. Afèche–Liu–Maglaras-style "scarce–intermediate–ample" language is appropriate for these regimes; one should not call the re-optimized value envelopes globally single-peaked until branch switching and rider composition are controlled.

7 A benchmark theorem, a full-model target, and the boundary

Takeaway. The strict peak order is proved on a common Poisson-WPBE branch. For the fully re-optimized mechanism, the honest theorem target is an ordering of argmax sets under an explicit branch-stability condition.

Theorem (ordered peaks of nested rescue controls). Fix one active cutoff-WPBE and rider-continuation branch. Suppose first-window failure contributes e^{-rm} and the nested terminal mechanisms induce $0 \le \lambda_0 < \lambda_F < \lambda_E$, with $r + \lambda_0 > 0$. Then

$$\Delta_P^0 = e^{-rm}(e^{-\lambda_0 m} - e^{-\lambda_F m}), \qquad \Delta_S^0 = e^{-rm}(e^{-\lambda_F m} - e^{-\lambda_E m})$$

are strictly single-peaked, with

$$m_P^* = \frac{\log[(r+\lambda_F)/(r+\lambda_0)]}{\lambda_F - \lambda_0}, \qquad m_S^* = \frac{\log[(r+\lambda_E)/(r+\lambda_F)]}{\lambda_E - \lambda_F},$$

and

$$\boxed{m_S^* < 1/(r+\lambda_F) < m_P^*}. \tag{17}$$

7.1 Proof and full-model lift

For $g_{a,b}(m) = e^{-am} - e^{-bm}$, $0 < a < b$, the derivative changes sign once and

$$m^*(a,b) = \frac{\log(b/a)}{b-a}, \qquad \frac{1}{b} < m^*(a,b) < \frac{1}{a}.$$

Apply this to the adjacent pairs $(r+\lambda_0, r+\lambda_F)$ and $(r+\lambda_F, r+\lambda_E)$.

For the optimized model define

$$\mathcal{S} = \arg\max_m (V_E - V_F), \qquad \mathcal{P} = \arg\max_m (V_F - V_0).$$

The desired statement is $\sup \mathcal{S} < \inf \mathcal{P}$. A sufficient condition is that both optimized gaps remain uniformly within half of their benchmark separation margins at $\bar{m} = 1/(r+\lambda_F)$. This condition permits multiple optimal policies; it does not assume differentiable envelopes.

7.2 Adversarial boundary

The common branch is essential. Let $\Delta_P = e^{-m} - e^{-2m}$ and multiply the search gain by the mechanism-specific continuation factor $\eta(m) = (1-e^{-m})^2$. Then

$$m_P^* = \log 2, \qquad m_S^* = \log(5/2) > m_P^*.$$

Both gains remain smooth and uniquely peaked. Rider-composition changes alone can reverse their order; policy switching can additionally create multiple peaks.

7.3 Closest theorem styles

- Afèche–Liu–Maglaras: scarce/moderate/ample thresholds and zero–positive–zero control value, not a unique peak.
- Zhao–Papier–Teo: quasi-convex thickness cost and an intermediate optimum; weak quasi-convexity does not imply uniqueness.
- Wang–Zhang–Zhang: characterize search scope by a scalar index crossing.
- Here: order the peaks of two *adjacent nested controls* after the inner cutoff-WPBE is made explicit.

Economic closure for search. On a common branch, let added scope create willing rate $\sigma(s)$ and contact rate $n(s)$. Under $J_\kappa = BM - \kappa Q^O$, an interior scope satisfies

$$\Psi(s;m) = \frac{Be^{-[\lambda_F+\sigma(s)]m}\sigma'(s)}{\kappa n'(s)} = 1.$$

Expand for $\Psi > 1$ and contract for $\Psi < 1$. With $\kappa = 0$, the solution is governed by the search cap or physical saturation, so it should be reported as the maximal-completion benchmark.

References. Afèche, Liu, Maglaras (2023), *M&SOM* 25(5), 1890–1908. Zhao, Papier, Teo (2024), *M&SOM*. Hu, Hu, Zhu (2022), *M&SOM* 24(1), 91–109. Wang, Zhang, Zhang (2024), *Operations Research* 72(3), 1278–1297.